

Heat Exchanger Lmtd Calculation

Page 1: Derivation of LMTD and Area Sizing

Heat exchangers are sized using the fundamental heat transfer equation: $Q = U * A * F * \text{LMTD}$. Q is the heat duty (kcal/h or Watts), U is the overall heat transfer coefficient ($\text{W/m}^2\text{-K}$), A is the heat transfer area (m^2), F is the configuration correction factor, and LMTD is the Log Mean Temperature Difference.

For counter-current flow, LMTD is derived as: $\text{LMTD} = (dT_1 - dT_2) / \ln(dT_1 / dT_2)$, where $dT_1 = T_{h_in} - T_{c_out}$ (temperature difference at hot fluid inlet) and $dT_2 = T_{h_out} - T_{c_in}$ (difference at hot fluid outlet). For co-current flow, $dT_1 = T_{h_in} - T_{c_in}$ and $dT_2 = T_{h_out} - T_{c_out}$. F is 1.0 for pure counter-current flow and less than 1.0 for multi-pass shell-and-tube exchangers.